

MATTER-FIRST AUTOMATED DISCOVERY OF TRANSIENT WARP PRECURSORS

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We build an automated pipeline that searches for warp-precursor spacetimes by proposing matter, solving the Einstein constraints, evolving the spacetime in full 3+1 numerical relativity, and testing whether null rays between fixed asymptotic endpoints arrive earlier than in flat space. MAP-Elites explores a 39-dimensional family of bicomplex-scalar Q-ball shells; **GR**Tresna solves the constraints; **GR**Teclyn evolves on GPUs; and a four-dimensional evolving null-geodesic tracer measures the fractional travel-time advantage f_{geo} . MAP-Elites saturates at candidate 87 ($S = 825.19$); CMA-ES warm-started from that elite raises the search score to 852.15 (candidate 146, $\Delta S \approx +27$). The improvement is confinement-gated persistence, not a larger raw geodesic excess. In the headline validation run ($L = 128$, $N = 256$, three AMR levels, $t = 30$), light arrives 20.2% earlier than the matched flat path, with all five rays reaching and maximum relative null-Hamiltonian drift 2.19×10^{-3} . The resolution ladder (RC/RM/RF at $N = 192/256/384$) yields 20.4%/20.2%/20.2%, with fine-medium relative difference $\ll 1\%$. Frozen-peak f_{geo} reaches $\sim 29.5\%$. Constraint norms stay bounded, but matter confinement falls to about 11% by $t = 30$. The evolution uses a bicomplex scalar model that preserves per-lump canonical/phantom signs through both the constraint solve and the dynamical stress tensor, together with an external trajectory controller that is switched off at $t = 4$. Because the headline geodesic is launched at $t_{\text{emit}} = 4$ —after pump-off—the 20.2% transit advantage is measured in a freely evolving Einstein–Klein–Gordon region. We therefore claim a numerically robust *transient* shortcut after a controlled ignition phase—not a passenger drive or a permanently maintained warp bubble.

I. INTRODUCTION

The goal of this work is an *automatic discovery pipeline* for faster-than-light (FTL) warp precursors. We define a warp-precursor as a constraint-satisfying matter configuration that generates a transient, effective null timing advantage prior to its own dispersal: null rays between prescribed asymptotic endpoints traverse the curved spacetime in less coordinate time than the matched flat-space path. Rather than hand-crafting a metric and hoping the implied stress-energy is physical, we want a closed loop that proposes candidates, evolves them under the Einstein equations, and retains only those that pass a gauge-invariant transport test.

The standard route is *geometry-first* (equivalently metric-first): one prescribes $g_{\mu\nu}$, reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and studies frozen slices [3]. That recipe is poorly suited to automated search. The implied matter is rarely the output of a consistent initial-value problem, and under dynamical evolution it typically disperses or collapses, destroying the intended channel. For a quality-diversity optimizer that must evaluate hundreds of candidates, a geometry-first genome is especially hostile: most deformations of $g_{\mu\nu}$ either violate the constraints, have no local matter realization, or fail as soon as time advances—so the archive fills with numerical failures rather than competing mechanisms.

We therefore invert the loop (Fig. 1). The optimizer proposes matter; **GR**Tresna returns constraint-

compatible initial data; **GR**Teclyn evolves the spacetime; and probes ask whether a null shortcut appears. In this paper “FTL” and “null shortcut” mean the same operational claim: an *effective* reduction of null travel time relative to flat space between the same asymptotic endpoints—never local superluminal motion outside the light cone. The fractional advantage is f_{geo} [Eq. (7)]; for example $f_{\text{geo}} = 0.20$ means the curved transit is 20% shorter than the flat comparison.

That criterion is load-bearing, not cosmetic. A warp precursor is useful only if it changes what light can do between fixed distant observers. Local coordinate signal speeds, lapse/shift cosmetics, or a positive frozen-slice diagnostic can look “superluminal” while no ray actually connects the endpoints sooner once the metric evolves. Asking whether null geodesics arrive earlier than in flat space is therefore the operational definition of the search target: a gauge-invariant transport certificate that the optimizer must earn, not a post-hoc plot.

Equally important is the *scoring metric* that turns each evolution into the fitness $S(\theta)$ used by the archive [Eq. (3)]. MAP-Elites converges to whatever S rewards. In earlier campaigns with weaker objectives the search reliably *cheated*: it harvested coordinate proxies while matter dispersed (“disperse-then-warp”), banked frozen-slice positives that failed the evolving geodesic test, or climbed scores through numerical pathologies—early horizons, integrator crashes, and other simulation-failure modes that accidentally produced large proxy signals. The present objective therefore rewards only the evolving 4D shortcut and persistence, gates health terms by nontriviality, penalizes exotic excess and horizon formation, and

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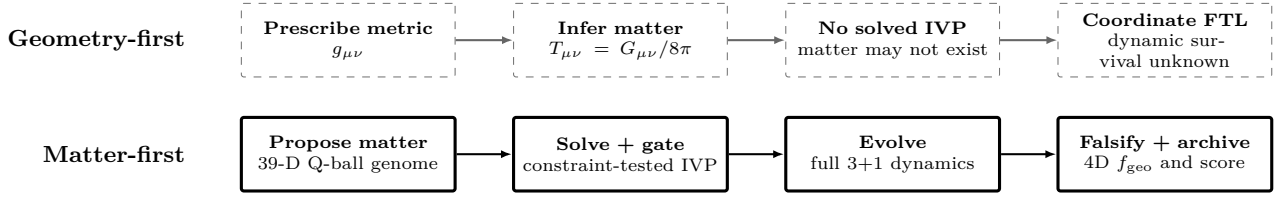


FIG. 1. Geometry-first versus matter-first discovery. Dashed: prescribe a metric and infer its stress-energy. Solid: propose matter, solve and test an initial-value problem (IVP), evolve it, and archive only candidates that pass the 4D null-transport test.

dispersion-gates transport credit so a flying-apart cloud cannot buy a high score. Without that metric design the archive does not discover physics; it discovers loopholes.

The search engine is MAP-Elites [7], a quality-diversity algorithm that keeps one elite per cell of a behavior archive rather than a single global optimum. Here the archive axes are peak shortcut strength and FTL life-time, so distinct transport mechanisms can coexist—but only among candidates that already cleared the scored falsification gate. We prefer this over a pure CMA-ES hill-climb, which collapses onto one basin and discards diverse near-misses, and over reinforcement learning, which would require dense credit assignment across expensive GPU evolutions with sparse, delayed rewards. After MAP-Elites identifies a high-scoring basin, CMA-ES warm-starts from the archive champion and locally refines the genome before any high-resolution replay or resolution ladder. In the present campaign, MAP-Elites saturates at candidate 87 ($S = 825.19$) and CMA-ES raises the score to 852.15 (candidate 146), a $\sim 3.3\%$ relative lift driven by improved confinement-gated persistence rather than a larger raw shortcut. Both discovery stages run on a deliberately coarse GRTEclyn grid (L , N , `max_level`) chosen for evaluation throughput; production claims are deferred to the later HQ / resolution matrix (Sec. II G).

A. Methodological Rigor

Three pillars distinguish this pipeline from geometry-first studies:

1. **Initial value problem:** constraint-clean data from physical scalar sources, not coordinate scripts.
2. **Dynamic feedback:** full BSSN/CCZ4 evolution; dispersing matter collapses the channel.
3. **4D geodesic falsification:** the headline diagnostic is `ftl_geo_evolving`—null rays integrated through the *time-evolving* metric stack. Frozen-slice geodesics are retained as diagnostics but are not the transport certificate.

B. Contributions

- (i) An end-to-end matter-first discovery pipeline—MAP-Elites proposal, CMA-ES local refinement of the archive champion, constraint solve, GPU evolution, and matched evolving-geodesic scoring—with an explicit reward rule applied to every evaluation (Secs. II–III).
- (ii) A hierarchical campaign protocol: quality-diversity search, CMA-ES hill-climb on the top genome, then longer higher-resolution replay and a three-run resolution ladder (RC/RM/RF) only on the refined champion (Sec. V).
- (iii) Explicit numerical limits of the surviving candidates: dispersing exotic matter and a transient ignition-phase controller—not passenger transport (Sec. VI).

II. MATTER-FIRST PIPELINE

A. Two-code toolchain

Two open numerical-relativity codes carry the loop. **GRTesna** is a constraint solver: given a matter configuration it returns constraint-satisfying initial data (γ_{ij} , K_{ij} , ϕ , Π) on the initial slice, so that the Hamiltonian and momentum constraints hold before any evolution. **GRTEclyn** [8, 9] is a GPU-accelerated, AMReX-based code that evolves those data forward with the CCZ4 formulation of the Einstein equations under moving-puncture gauge. Neither code is told what metric to produce; the geometry is whatever the proposed matter sources.

B. Search Loop

One evaluation maps a parameter vector θ to a scalar fitness: $\theta \rightarrow \text{initial data} \rightarrow \text{GRTEclyn} \rightarrow \text{diagnostics} \rightarrow S(\theta)$. The optimizer proposes θ , **GRTesna** solves the constraints, **GRTEclyn** evolves the spacetime on GPU, and post-processing probes reduce the evolution to the fitness $S(\theta)$ that steers the next proposal.

The matter is a canonical complex scalar field ϕ with conjugate momentum Π and potential $V(\phi)$. In the 3+1 split, a scalar field contributes an energy density ρ and a momentum density S_i that act as the sources in the Hamiltonian and momentum constraints,

$$\rho = \frac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \frac{1}{2}\Pi^2 + V(\phi), \quad S_i = -\Pi\partial_i\phi, \quad (1)$$

where ψ is the conformal factor and $\tilde{\nabla}$ the conformal spatial gradient. The three terms in ρ are, in order, the gradient (spatial shape), kinetic (time variation), and potential (rest) energy of the field; S_i is nonzero only when the field oscillates in time ($\Pi \neq 0$) with a spatial gradient, which is what lets a moving scalar lump carry momentum and drive a shift. Equation (1) is therefore the precise statement of “the matter tells the geometry what to be”: θ sets ϕ and Π , and **GRTresna** returns the (γ_{ij}, K_{ij}) consistent with these sources.

The word “canonical” describes the Klein–Gordon equation of motion, not the sign with which every initial lump gravitates. For each lump the optimizer stores $x_{\text{ex},k} \in [0, 1]$, but configuration construction rounds it to $e_k = \text{round}(x_{\text{ex},k})$ and sets $s_k = (-1)^{e_k}$. Thus there is no physical fractional interpolation. In the **GRTresna** constraint solve the signed independent-lump approximation is

$$\rho_{\text{ID}} = \sum_k s_k \rho_k, \quad (S_i)_{\text{ID}} = - \sum_k s_k \sum_{A=1}^2 \Pi_{A,k} \partial_i \phi_{A,k}, \quad (2)$$

with inter-lump stress-tensor cross terms omitted. The profiles themselves are then superposed into two painted complex fields—one canonical, one phantom—by the bicomplex model. Candidate 146 rounds to $s_k = (+1, -1, -1, +1, -1)$: two canonical and three phantom sources, with these signs preserved in both the constraint solve and the dynamical evolution.

C. Q-ball building blocks

The matter is built from Q-balls, and this choice was forced on us by failure. Earlier ansatz families—spherical-harmonic scalar shells and free massive (Klein–Gordon) wave packets—have no bound state and simply spread: their structural persistence collapses to a few percent well before the end of the evolution, taking any transient shortcut with them (Sec. VIA). A Q-ball is a localized, non-topological soliton of a complex scalar field: the field carries a conserved Noether charge under a global phase symmetry, and that charge holds a compact lump of field together against dispersal. This self-binding is precisely what the earlier families lacked. We choose a *complex* rather than real scalar for two reasons. First, only a complex field has the phase symmetry that supports a stable charged lump. Second, its harmonic phase rotation $\phi \propto e^{-i\omega t}$ makes $\Pi = \dot{\phi} \neq 0$, so by Eq. (1) the lump carries a momentum density S_i and can source a nontrivial shift—the ingredient a transport channel needs. Each

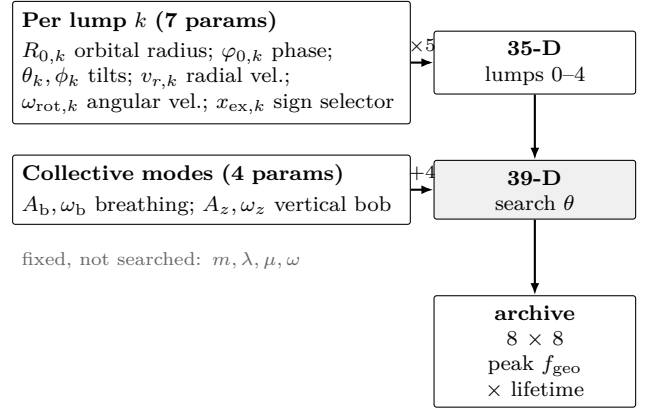


FIG. 2. Search parameterization of the trajectory Q-ball shell: $5 \times 7 = 35$ per-lump placement parameters plus 4 collective modes give the 39-D genome θ . MAP-Elites stores one elite per cell of an 8×8 archive indexed by peak shortcut and FTL lifetime. Q-ball microphysics is held fixed.

lump starts from a spherically symmetric equilibrium profile obtained by solving the radial ODE for the chosen sextic potential ($m = 1, \lambda = 640, \mu = 85333, \omega = 0.8$); these fixed microphysical constants are not searched.

D. Shell Parameterization

The search vector has 39 dimensions because the arrangement of the lumps, not the lump microphysics, is what is optimized. Five lumps each carry seven searchable placement parameters—orbital radius R_0 , phase, two tilt angles, radial velocity, angular (rotation) velocity, and a source-sign selector—giving $5 \times 7 = 35$ per-lump dimensions. Four collective variables set the shell’s breathing amplitude and frequency and its vertical oscillation amplitude and frequency, for $35 + 4 = 39$ in total (Fig. 2). Although sampled continuously, the sign variable is applied as the binary selector above; it lets the optimizer choose which initial-data lumps source positive or negative stress-energy. MAP-Elites [7] keeps one elite per cell of an 8×8 behavior archive indexed by peak shortcut strength and FTL lifetime, which preserves distinct mechanism families rather than collapsing onto a single optimum.

E. Constraint gate and source-model consistency

A post-load constraint gate rejects inconsistent initial data before any GPU time is spent: the **GRTresna** Hamiltonian and momentum residuals must fall below 5%, otherwise the candidate is discarded and logged. The campaign uses the **grtresna_bicomplex_scalar** matter model, which carries two independent complex scalar fields—one canonical, one phantom—so that the per-lump gravitational signs in Eq. (2) are preserved in both

the **GRtresna** constraint solve and the **GRteclyn** evolution stress tensor $T_{\mu\nu}$. This eliminates the initial-data / evolution sign mismatch that afflicted earlier single-field campaigns. The post-load Hamiltonian and momentum gate, together with bounded CCZ4 constraint growth during evolution, therefore establishes both numerical control and source-model consistency for the bicomplex sector.

F. Scoring: how every evaluation is rewarded

Every evaluation in the campaign is reduced to one fitness value $S(\theta)$ by the same fixed rule, so candidates are ranked consistently. Each named quantity below is a normalized diagnostic in $[0, 1]$ (penalties are negative); the score is their weighted sum,

$$S = 1000 f_{\text{geo}}^{\text{evol}} + 1000 f_{\text{geo}}^{\text{op}} + 600 P_{\text{ftl}} + 200 F_{\text{op}} + 200 C_{\text{curv}} + 5 G_{\text{nt}} + g_{\text{h}} \Sigma_{\text{health}} + 40 w_x X_{\text{exotic}} + 200 H_{\text{hor}}, \quad (3)$$

with the health block

$$\Sigma_{\text{health}} = 150 \text{ surv} + 60 E + 40 \text{ stab} + 30 \text{ cmv} + 10 \mathcal{H}_{\text{ok}} + 15 I, \quad (4)$$

gated by the nontriviality factor $g_{\text{h}} \in [0, 1]$ so that a trivial (flat) configuration earns no health reward. Table I lists what each term measures. Two design choices matter. First, transport dominates: the evolving 4D shortcut $f_{\text{geo}}^{\text{evol}}$ and its geodesic counterpart carry the largest weights, so the score chiefly tracks genuine null-transport gain. Second, the persistence and operational terms $P_{\text{ftl}}, F_{\text{op}}$ are *dispersion-gated*—multiplied by the fraction of matter still confined—so a candidate cannot bank a co-ordinate superluminal channel after its lumps have flown apart. The exotic penalty $X_{\text{exotic}} < 0$ charges for energy-condition violation at campaign weight w_x . All single scores quoted later are values of this S on a $t = 16$ evolution: search rankings, not physical observables, and never compared across runs of different duration.

For the CMA-ES champion (candidate 146) the search-tier score is $S = 852.15$, dominated by evolving-shortcut and gated-persistence terms. The CMA-ES refinement lifted the score from the MAP-Elites plateau ($S = 825.19$, candidate 87) primarily through improved confinement-gated fitness ($0.396 \rightarrow 0.417$), not a larger raw geodesic excess (the search-tier f_{geo} actually decreased from 25.3% to 20.1%). This illustrates the dispersion gate at work: CMA-ES found a genome that trades a smaller instantaneous shortcut for better structural persistence.

G. Search and validation stages

MAP-Elites and CMA-ES evaluate $\mathcal{O}(10^2)$ genomes, so the discovery loop uses a reduced GRteclyn

TABLE I. Reward terms in the campaign objective [Eq. (3)], applied identically to every MAP-Elites evaluation.

symbol	rewards / penalizes	weight
$f_{\text{geo}}^{\text{evol}}$	4D evolving null shortcut	1000
$f_{\text{geo}}^{\text{op}}$	geodesic (frozen) shortcut	1000
P_{ftl}	FTL persistence (gated)	600
F_{op}	operational coord. FTL (gated)	200
C_{curv}	curvature activity	200
G_{nt}	nontrivial geometry	5
surv	structural survival	150 g_{h}
E	energy-condition health	60 g_{h}
stab	metric stability	40 g_{h}
cmv	co-moving stability	30 g_{h}
I	instability penalty	15 g_{h}
$\mathcal{H}_{\text{ok}}$	constraint health	10 g_{h}
X_{exotic}	exotic-matter penalty	40 w_x
H_{hor}	horizon-formation penalty	200

evolution configuration for wall-clock speed. Campaign env variables **GRtresna_evolution_L_FULL**, **GRtresna_evolution_N_FULL**, **GRtresna_evolution_max_level**, and **GRtresna_evolution_regrid_threshold** set the search-tier AMReX inputs **max_level** and **regrid_threshold** together with the evolution box (L, N) and t_{stop} . Archive / CMA scores are therefore *search-tier* diagnostics, not continuum claims. Only the frozen champion is re-run at production resolution for validation.

- **Stage 1 — MAP-Elites discovery (fast search grid):** quality-diversity search at $L = 64$, $N = 128$ ($h = 0.5$), **max_level**=1, **regrid_threshold**=0.1, and $t_{\text{stop}} = 16$ (bicomplex matter; per-lump gravitational signs retained in evolution). Coarse AMR is intentional: it buys enough evaluations to fill the behavior archive. Produces a diverse archive and a provisional score leader.
- **Stage 2 — CMA-ES refinement (same search grid):** warm-start from the MAP-Elites champion (and optionally its nearest elites); local covariance-matrix adaptation at the *same* $(L, N, \text{max_level}, t_{\text{stop}})$ and scoring rule. Only the refined top genome is promoted further.
- **Stage 3 — production replay (HQ):** the frozen CMA-ES champion is re-solved and evolved at higher fidelity / longer t_{stop} —typically $L = 128$, $N = 256$, **max_level**=3, **regrid_threshold**=0.02, $t_{\text{stop}} = 30$ —with multiple emission times and multi-ray geodesic probes. Paper headline numbers come from this stage, not from MAP-Elites.
- **Stage 4 — resolution matrix:** matched coarse/medium/fine resolution runs (RC/RM/RF

TABLE II. GRTEclyn evolution settings: search tier (MAP-Elites / CMA-ES) versus production validation (HQ / Richardson medium). Search uses a reduced grid for throughput; validation upgrades L , N , max_level , and t_{stop} . Campaign env: `GRTRESNA_EVOLUTION_{L,N}_FULL`, `_MAX_LEVEL`, `_REGRID_THRESHOLD`.

parameter	search	HQ / RM
L	64	128
N	128	256
$h = L/N$	0.5	0.5
max_level	1	3
regrid_threshold	0.1	0.02
t_{stop}	16	30

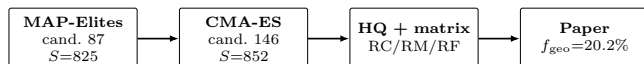


FIG. 3. Outer campaign ladder. MAP-Elites finds the diverse archive and champion (87); CMA-ES refines it to candidate 146; only the refined champion enters high-resolution replay and the resolution matrix.

at fixed $L = 128$, $N = 192/256/384$) test whether the shortcut survives discretization changes at the production max_level . Domain-ladder and wave-form runs are deferred.

III. SEARCH METHODOLOGY

A. MAP-Elites quality-diversity

MAP-Elites [7] is a quality-diversity search: instead of returning one champion, it maintains an 8×8 archive with at most one elite per behavior cell. A candidate displaces the occupant of its cell only if its score $S(\theta)$ [Eq. (3)] is higher. The two archive axes are peak shortcut strength and FTL lifetime, so the campaign can keep short-lived strong channels and longer weaker ones side by side. The bicomplex campaign (`qball_traj_bicomplex_v1`) scheduled 200 evaluations: 148 completed GPU evolution, 50 failed the post-load gate, one failed the GRTresna admission gate, and one failed the constraint solve. Nine cells were occupied (14.06% coverage). All nontrivial elites lie in the highest lifetime row, while peak strength spans the archive. The best score rose from 690.1 to 825.19 and last improved at batch 10, when candidate 87 entered cell (7, 7).

B. Archive outcome and CMA-ES refinement

The bicomplex MAP-Elites winner (candidate 87, $S = 825.19$) combines a search-tier evolving shortcut of $f_{\text{geo}} =$

TABLE III. Two-stage search hand-off: MAP-Elites champion (87) $\rightarrow$ CMA-ES refined champion (146) $\rightarrow$ HQ validation (RM). Search-tier grid: $L=64$, $N=128$, $\text{max_level}=1$, $t=16$. HQ grid: Table II.

stage	S	rays	peak f_{geo}	gated fitness
MAP-Elites (87)	825.19	3/3	25.35%	0.396
CMA-ES (146)	852.15	3/3	20.12%	0.417
HQ RM (146)	—	5/5	20.21%	—

25.3% with dispersion-gated fitness 0.396, constraint-growth score 0.983, and confinement retention 0.396. CMA-ES warm-started from this genome and ran 150 evaluations (144 GPU-ok) at the same search grid, raising the score to 852.15 (candidate 146, $\Delta S \approx +27$). The improvement is not a bigger raw shortcut—the search-tier f_{geo} actually *decreased* from 25.3% to 20.1%—but a better confinement-gated persistence (0.396 $\rightarrow$ 0.417). CMA-ES converged to a genome with nearly static lump placement: very slow angular velocities and small breathing, producing a quasi-static scaffold that keeps matter confined longer (Fig. 7).

Search-tier confinement already falls to $\sim 40\%$ by $t = 16$, so longer production runs are expected to stress structural persistence even if the evolving null shortcut remains positive.

IV. DIAGNOSTICS AND PROBE VALIDATION

A. The 4D evolving null-geodesic trace

The transport claim rests on tracing light rays through the simulated spacetime, not on reading coordinate speeds off a slice. A coordinate operational shortcut F_{op} [Eq. (6)] can flag tilted light cones, but it is partly gauge dependent and can be inflated by a coordinate choice. The authoritative probe therefore integrates the null-geodesic equations

$$\dot{x}^\mu = g^{\mu\nu} k_\nu, \quad \dot{k}_\mu = -\frac{1}{2}(\partial_\mu g^{\alpha\beta}) k_\alpha k_\beta, \quad (5)$$

for a photon with position x^μ and momentum k_μ . “4D” means the ray is not confined to one snapshot: it is pushed through a stack of stored time slices, so at each step the metric $g_{\mu\nu}(t, \mathbf{x})$ used is the one that actually exists when the photon is at that point. This distinguishes it from a *frozen* trace, which wrongly assumes a single slice persists for the whole flight.

The construction is as follows. A ray is emitted at time t_{emit} from a fixed source point toward a fixed receiver. The quantities that enter the shortcut formulas are *transit durations*, not absolute grid arrival times: t_B^{curved} (respectively t_{min}) is the coordinate time elapsed along the curved path after emission, and t_B^{flat} (respectively t_{flat}) is the matched flat-space flight duration between the same endpoints. Absolute arrival on the grid

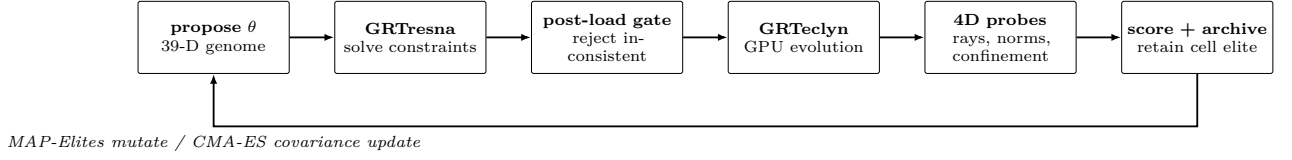


FIG. 4. Inner evaluation loop shared by MAP-Elites and CMA-ES. Failed constraint solves and post-load checks never enter GPU evolution; successful evaluations compete for archive cells (MAP-Elites) or update the CMA distribution.

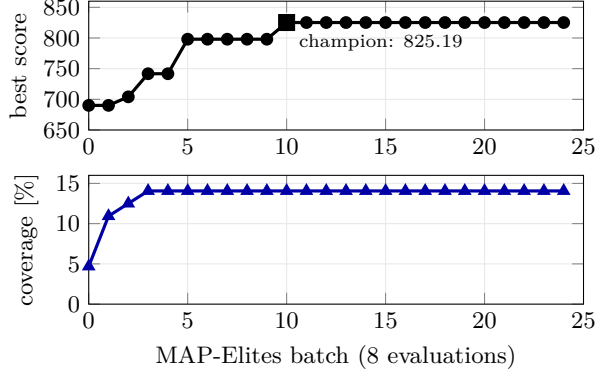


FIG. 5. MAP-Elites search progress over 200 evaluations (qball_traj_bicomplex_v1). Top: running best archive score. Bottom: fraction of occupied behavior cells. The final champion (candidate 87) enters cell (7,7) at batch 10 with score 825.19. Curves are read from the stored campaign history.

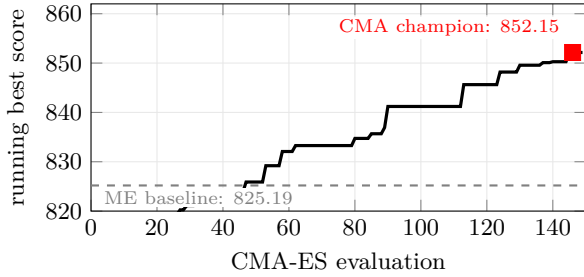


FIG. 6. CMA-ES refinement progress over 150 evaluations (qball_traj_bicomplex_cmaes_v1), warm-started from MAP-Elites champion 87. Dashed line: MAP-Elites best (825.19). The running best exceeds the ME baseline early and reaches 852.15 at eval 146.

is $t_{\text{arrival}} = t_{\text{emit}} + t_B^{\text{curved}}$. The fractional advantage is

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \quad (6)$$

$$f_{\text{geo}} = \max\left(0, \frac{t_{\text{flat}} - t_{\text{min}}}{t_{\text{flat}}}\right), \quad (7)$$

so $f_{\text{geo}} = 0$ means no shortcut and $f_{\text{geo}} = 0.20$ means the curved transit is 20% shorter than the flat comparison. A launch is accepted only if its rays stay physical: the

relative drift in the null Hamiltonian $g^{\mu\nu}k_\mu k_\nu = 0$ must remain below 10^{-2} , which rejects rays that have accumulated interpolation error. The simulation itself runs to $t = 30$, but that is the evolution horizon of the metric stack, not the emission schedule. A ray launched at t_{emit} still needs a long remaining segment of evolved geometry to reach the far endpoint (flat transit is $t_{\text{flat}} = 14.4$), so late launches near $t = 30$ cannot finish a fair comparison. We therefore sweep seven early launch times, $t_{\text{emit}} = 0, 2, \dots, 12$ (five rays each), and report the largest accepted f_{geo} —the best moment to “surf” the channel. Across the validation matrix the peak sits at $t_{\text{emit}} = 4$ (just after the PD pump switches off) and has already fallen by $t_{\text{emit}} = 12$, so the window covers the opening of the channel rather than cutting it off.

B. Probe controls

Prescribed Alcubierre [4] ($f_{\text{geo}} \approx 32\%$) and Minkowski ($f_{\text{geo}} = 0$) calibrate the codebase. A previously identified negative control, **eval 086**, has a positive frozen-slice diagnostic but fails the evolving HQ trace. The CMA-ES champion (candidate 146) supplies the positive case: every accepted launch in all three resolution-matrix runs has 5/5 rays reaching and passes the null-Hamiltonian gate.

V. RESULTS: CANDIDATE-146 WARP PRECURSOR

A. Provenance and genome

Candidate 146 is the CMA-ES-refined champion promoted from the bicomplex MAP-Elites campaign. MAP-Elites saturated at candidate 87 ($S = 825.19$); CMA-ES warm-started from that genome and after 150 evaluations raised the score to $S = 852.15$ (candidate 146, $\Delta S \approx +27$). Its provenance freezes the exact 39-dimensional genome, source checksums, solver parameters, and source commit. The configuration uses the `gtrtesna_bicomplex_scalar` matter model with per-lump signs preserved through both the constraint solve and the GPU evolution. The resolution-matrix RC and RM runs were metrics-only; the RF (fine) run was evolved with `GRTECLYN_FRAMES=1` and fixed color scales, from which field-map movies were produced.

B. Matter dynamics

The five Q-balls do not share one rigid orbit. Lump k moves in its own tilted orbital plane, with the support center prescribed by

$$\begin{aligned} r_k(t) &= \max[r_{\min}, R_{0,k} + v_{r,k}t + A_b \sin(\omega_b t)], \\ \varphi_k(t) &= \varphi_{0,k} + \omega_{\text{rot},k}t, \quad z_b(t) = A_z \sin(\omega_z t), \end{aligned} \quad (8)$$

followed by rotation into the laboratory frame through the two tilt angles. Thus each lump spirals radially, rotates in a different plane, and shares a collective radial “breathing” and vertical oscillation. For candidate 146, $A_b = 0.083$, $\omega_b = 0.429$, $A_z = 2.32$, and $\omega_z = 0.016$. Breathing remains similarly small in the MAP-Elites champion (87 had $A_b \approx 0.078$); what CMA-ES changed is the orbital drive. The initial radii are $R_0 = (3.69, 5.50, 2.83, 5.59, 2.61)$; all angular velocities are negative, $\omega_{\text{rot}} = (-0.0017, -0.0007, -0.0031, -0.0009, -0.0049)$, so all five trajectories are retrograde but extremely slow. Radial drifts are small [$v_r = (-0.060, -0.010, 0.001, -0.005, -0.006)$]; the slow rotation and drift—roughly an order of magnitude quieter than candidate 87—is what CMA-ES converged to in order to maximize confinement-gated persistence. Lumps 0 and 3 are canonical; lumps 1, 2, and 4 are phantom after rounding.

Figure 7 shows these prescribed support-center trajectories at four times. The Q-ball fields are initially boosted along the corresponding motion. During evolution, a localized proportional-derivative (PD) controller drives the two real components of the complex field toward a moving target profile:

$$\begin{aligned} \partial_t \Pi_A|_{\text{PD}} &= -wk_p(\phi_A - \phi_A^*) \\ &\quad - wk_d(\Pi_A - \Pi_A^*), \quad A = 1, 2, \end{aligned} \quad (9)$$

with $k_p = 12$, $k_d = 7$, and w the product of a localized envelope and a constraint-sensitive governor. This is the precise meaning of the “moving well.” It is an external, non-Lagrangian source in the scalar momentum equations; no corresponding pump stress-energy is added to $T_{\mu\nu}$. Therefore $\nabla_\mu T^{\mu\nu} = 0$ is not exact and the contracted Bianchi identity is not satisfied by a closed matter system.

The PD controller is active only during the early ignition phase: the pump switches off at $t_{\text{pump}} = 4$ (`rl_pump_stop_time`). After that the matter evolves freely under Einstein–Klein–Gordon dynamics with the bicomplex stress-energy. The bounded constraints ($\|\mathcal{H}\|_2 \leq 4.73 \times 10^{-3}$, $\|\mathcal{M}\|_2 \leq 9.16 \times 10^{-4}$ in RM) show that the evolution remains numerically controlled. The support centers are the intended dynamical scaffold, not tracked density isosurfaces. The field follows them imperfectly: confinement falls from 72% to 11% and rms spread grows by a factor 3.27 by $t = 30$. Candidate 146 is therefore a transient, initially driven configuration, not a bound five-body soliton.

TABLE IV. Candidate-146 resolution matrix. All runs use $L = 128$, three AMR levels (`max_level` = 3, `regrid_threshold` = 0.02), and stop at $t = 30$. Peak f_{geo} is the best evolving 4D shortcut; frozen peak is the maximum of the frozen-slice f_{geo} time series.

run	N	h	peak f_{geo}	frozen peak	t_{emit}
RC	192	0.667	20.44%	29.15%	4
RM	256	0.500	20.21%	29.45%	4
RF	384	0.333	20.19%	29.54%	4

C. 4D falsification certificate

The headline claim is operational: in the RM run, null rays between the fixed endpoints arrive earlier than in flat space. The evolving-geodesic artifact reports:

peak 4D f_{geo}	20.21%
frozen-peak f_{geo}	29.45%
rays reached	5/5
t_{emit}	4.00
t_{arrival}	15.49
$t_{\text{flat}}^{\text{flat}}$ $t_{\text{transit}}^{\text{transit}}$	14.40
$h_{\text{quality_ok}}$	true ($h_{\text{rel}}^{\text{max}} = 2.19 \times 10^{-3}$)
search-tier peak (CMA-ES)	20.12% (3/3 rays)

Explicitly, $(t_{\text{flat}} - t_{\text{min}})/t_{\text{flat}} = (14.40 - 11.49)/14.40 = 20.21\%$, matching Eq. (7). The peak emission time $t_{\text{emit}} = 4$ coincides with the moment the PD pump switches off, suggesting that the transport channel opens as the driven ignition ceases and the matter begins its free evolution. The production HQ value agrees with the search-tier CMA-ES value to within 0.5% relative, confirming that the coarse search grid captures the essential physics. The endpoint timing corresponds to an effective $c/(1 - f_{\text{geo}}) \simeq 1.25c$ relative to the matched flat path; locally the ray remains null.

D. Resolution matrix

The completed validation matrix is a three-run resolution ladder at fixed domain $L = 128$: coarse RC ($N = 192$, $h = 0.667$), medium RM ($N = 256$, $h = 0.500$), and fine RF ($N = 384$, $h = 0.333$). RM is the headline HQ baseline; paper numbers come from this run. No domain ladder (DS/DL) was run for candidate 146; that check is deferred to future work.

Across the resolution ladder, the same genome shortens null travel by 20.19–20.44% relative to flat space. The fine–medium relative difference is $(20.21 - 20.19)/20.21 = 0.1\%$, far below the preregistered 10% matrix tolerance. The frozen-peak f_{geo} is similarly stable at $\sim 29.5\%$ across all three resolutions. For the $L = 128$ domain, the $t = 30$ stop precedes the arrival of boundary-reflected gauge modes at the central region: in 1+log slicing those

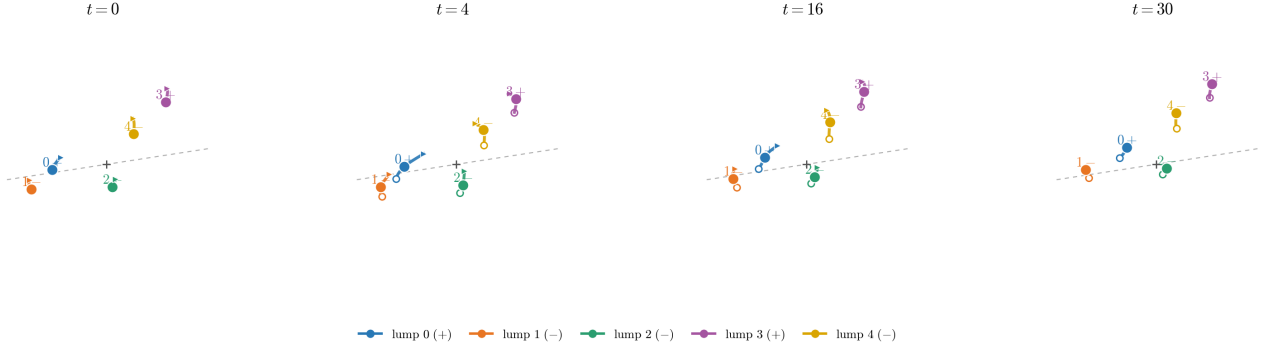


FIG. 7. Three-dimensional dynamics encoded by the frozen candidate-146 genome. Panels show the five prescribed PD-controller centers at $t = 0, 4, 16$, and 30 . Filled circles are the centers at the stated times. To make the small motion legible, each local trajectory glyph covers four code-time units before and after the snapshot (where available), with its displacement about the filled center magnified by $4\times$; the hollow circle and arrow mark the start and end of that window. Colors and labels identify the lumps (+ canonical, - phantom), and the dashed line marks the x -directed null-probe axis. The magnification is representational: the centers are plotted at their unscaled positions. CMA-ES converged to very slow angular velocities ($|\omega_{\text{rot}}| \lesssim 5 \times 10^{-3}$) and small breathing ($A_b = 0.083$), producing a quasi-static scaffold that maximizes confinement-gated persistence.

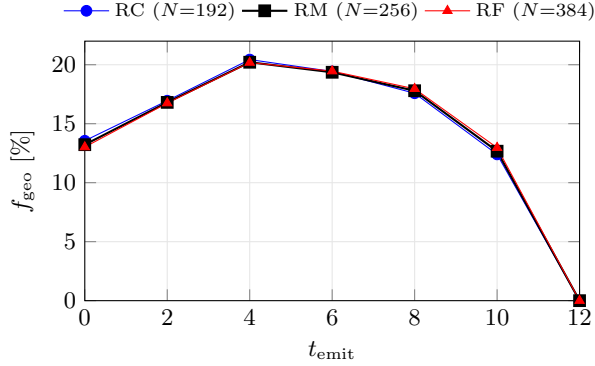


FIG. 8. Evolving null-geodesic emission sweeps ($t_{\text{emit}} = 0, 2, \dots, 12$) for all three resolution-matrix runs; the metric evolves to $t = 30$, but late launches lack enough remaining geometry for a complete transit. All runs peak at $t_{\text{emit}} = 4$, just after PD pump-off, and agree within 0.26 percentage points. Every point shown has 5/5 rays reaching and passes the null-Hamiltonian quality gate.

modes propagate at characteristic speeds up to $\sim\sqrt{2}c$, so a one-way crossing from the outer boundary requires $t \gtrsim L/(2\sqrt{2}) \approx 45$.

For unequal spacings, a Richardson estimate would assume

$$Q(h) = Q_0 + Ch^p, \quad \frac{Q_c - Q_m}{Q_m - Q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p}. \quad (10)$$

The sequence $20.44\% \rightarrow 20.21\% \rightarrow 20.19\%$ is monotonically decreasing, consistent with convergence. The fine-medium difference is so small (0.02% absolute) that the extracted convergence order would be dominated by numerical noise. We therefore accept continuum stability of the 20.2% result without quoting a formal extrapolated value or grid-convergence index.

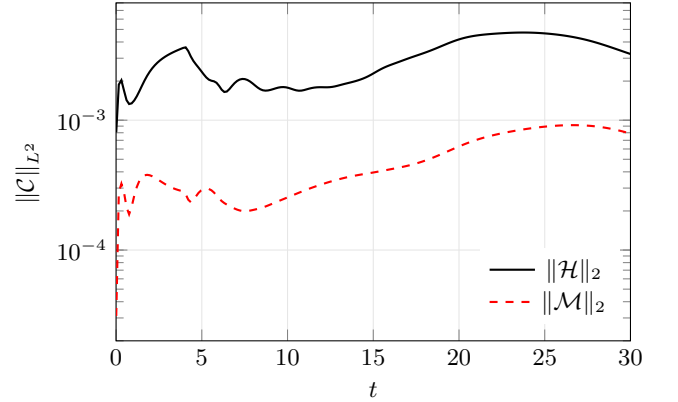


FIG. 9. Constraint norms in the headline RM run. The complete 3000-step diagnostic is plotted from a decimated table for typesetting only; maxima are evaluated from the full data.

E. Evolution constraints

For RM, the volume-weighted norms remain below $\|\mathcal{H}\|_2 = 4.73 \times 10^{-3}$ and $\|\mathcal{M}\|_2 = 9.16 \times 10^{-4}$ through $t = 30$ (Fig. 9). These are comparable to the search-tier values and remain bounded throughout the evolution. The constraint norms peak during the early driven phase ($t \lesssim 4$) and then plateau or slowly grow during the free evolution, consistent with the dispersing matter producing less active refinement at late times.

F. Frozen versus evolving shortcut, and confinement

RM's frozen-peak f_{geo} reaches 29.45%, compared with the evolving 4D sweep peak of 20.21%. The $\sim 9\%$ gap

is physical: the frozen trace assumes that one snapshot persists for the entire ray flight, whereas the 4D trace samples the actual changing metric and therefore “sees” the channel open and then weaken during propagation. Only the latter is used for the transport claim. The frozen peak is stable across all three resolutions (29.1–29.5%), confirming that the metric snapshot quality is not resolution-sensitive.

The matter structure is not durable. In RM, confined matter fraction falls from 72% at $t = 0$ to 11% by $t = 30$, while the rms spread grows by a factor 3.27. The peak shortcut at $t_{\text{emit}} = 4$ occurs just as the pump switches off, and the evolving f_{geo} curve falls steadily at later emission times. Successful ray transport therefore coexists with substantial dispersal: the channel is transient.

G. What moves faster than light

Nothing physical moves outside its local light cone. The quoted $\sim 20\%$ is a *null-geodesic timing advantage* through the evolved geometry relative to a flat reference path between the same endpoints. Coordinate light-speed maps visualize the tilted cones but do not establish the result by themselves.

VI. DISCUSSION

A. The dispersion problem

Candidate 146 should be read against the many configurations that did not work. Across roughly a dozen ansatz families and several full MAP-Elites campaigns we repeatedly found the same failure mode: the supporting matter disperses and the transport channel closes with it. Spherical-harmonic shells yielded a single weak hit in ~ 200 evaluations. Free massive scalar packets have no bound state and blow apart almost immediately. Bicomplex phantom lumps opened a gauge-invariant shortcut but lost $\sim 98\%$ of their structure by $t = 16$ (structural persistence ≈ 0.02). Boosted and stiffened Q-ball variants improved early confinement but still sheared apart once placed on an orbit. This is why the score explicitly gates transport by confinement (Sec. II F): without it, the optimizer is rewarded for a coordinate superluminal channel produced by matter that has already flown apart. CMA-ES improved confinement-gated persistence relative to the MAP-Elites champion ($0.396 \rightarrow 0.417$), but candidate 146 still retains only about 11% of its confined matter by $t = 30$ at HQ. The search-tier confinement ($\sim 40\%$ at $t = 16$) falls more steeply under the longer, higher-resolution production evolution. The honest summary is that a *transient* shortcut is reproducible and robust, while a *long-lived*, well-confined channel remains an open problem.

B. Interpretation

Candidate 146 demonstrates that matter-first search with 4D falsification can identify a constraint-gated, transiently driven evolution whose transport gain survives production resolution and refinement. The strongest claim is continuum stability of the measured null timing: the resolution ladder RC–RM–RF yields 20.4%/20.2%/20.2% with fine–medium relative difference $\ll 1\%$, and all ray-quality gates pass across all three runs. The bicomplex matter model preserves per-lump canonical/phantom signs through both the constraint solve and the evolution, so the sign-mismatch caveat of earlier single-field campaigns no longer applies. The PD controller is active only for $t < 4$; after pump-off, the system evolves under standard Einstein–Klein–Gordon dynamics with the bicomplex stress-energy.

C. Applications and next steps

The pipeline supports (i) inverse design of null transport under energy-condition penalties, (ii) QD atlases of mechanism families, and (iii) Pareto maps of shortcut versus exotic cost and persistence. Immediate priorities are to run a domain ladder (DS/DL) to verify boundary insensitivity, add a pump-work diagnostic to quantify the energy deposited during the ignition phase, and extend the evolution to longer t_{stop} to track late-time dispersal. Longer-term goals include optimizing for non-dispersing matter models and confinement world-tube objectives.

VII. CONCLUSION

We presented an automatic matter-first discovery pipeline: MAP-Elites proposes scalar configurations, CMA-ES locally refines the archive champion, the constraints are solved, the spacetime is evolved, and HQ 4D null geodesics ask whether light arrives earlier than in flat space. MAP-Elites saturated at candidate 87 ($S = 825.19$); CMA-ES improved the search score to 852.15 (candidate 146), a lift driven by confinement-gated persistence rather than a larger raw shortcut. The CMA-ES champion, a bicomplex five-lump Q-ball shell, yields $f_{\text{geo}} = 20.2\%$ in the headline RM run—light 20% earlier—and 20.2–20.4% across the three-run resolution ladder (fine–medium relative difference $\ll 1\%$). The frozen-peak f_{geo} reaches $\sim 29.5\%$, stable across resolutions. All rays pass quality gates (5/5 reaching; matrix-wide $h_{\text{rel}}^{\text{max}} \leq 3.8 \times 10^{-3}$, with 2.19×10^{-3} in the headline RM run). The bicomplex matter model preserves per-lump canonical/phantom signs through both the constraint solve and the evolution. The PD controller switches off at $t = 4$; because the headline geodesic launches at $t_{\text{emit}} = 4$, the 20.2% advantage is measured in freely evolving Einstein–Klein–Gordon dynamics. Confinement falls to $\sim 11\%$ by $t = 30$, so the shortcut is transient. Candidate 146

is therefore a reproducible, continuum-stable numerical warp precursor produced by an automated two-stage search, not a continuum-extrapolated or functional warp drive.

VIII. CODE AND DATA AVAILABILITY

The project repository contains the campaign configuration and archive, candidate-146 frozen provenance (under `runs/grtresna_promote/sources/`), the three completed resolution-matrix validation summaries (RC/RM/RF evolving geodesic artifacts, constraint

histories, and confinement timeseries). Figure files live in `research/neuralspacetime/figures/`; the article directory holds the reduced numerical tables used by TikZ/PGFPlots. MAP-Elites and CMA-ES trajectory files are under `runs/grtresna_qd/` and `runs/grtresna_cmaes/`.

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